Predicting Emotion in Music Using Audio Features

A Genre Stratified Machine Learning Study of Valence and Energy on the Spotify Tracks Dataset

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1. Introduction

Spotify has evolved from a simple music player into the world's most dominant audio ecosystem. Today, it is the world's most popular audio streaming service, offering over 100 million tracks, 7 million podcast titles, and 500,000 audiobooks. As of early 2026, Spotify operates in 184 markets with over 751 million monthly active users, including 290 million paying subscribers. It has an annual revenue of approx. $19.44 billion (up from 16.96$ billion in 2024). One of the key factors for its success is Spotify's recommendation engine. It is widely considered the gold standard in the industry. Its success lies in the fact that it doesn't just look at what you like, it looks at why you like it and who else shares your specific quirks. Spotify uses a version of James Russell's Circumplex Model of Affect. Instead of labeling a song with a single emotion like "sad", it plots every track on a 2D coordinate system. In this model, the valence of a song refers to its positive or negative sound, whereas arousal refers to the level of intensity in the music. Music emotion recognition is a key concept in music information retrieval and affective computing. With the popularity of streaming platforms such as Spotify, the relationship between music characteristics and emotional perception has become increasingly relevant. This information can be used in music recommendation systems, playlist generation, and understanding user preferences.

Spotify acquired The Echo Nest in 2014, a music intelligence company that provided the technological foundation for many of Spotify's audio analysis features, including valence Despite the utility of the VA model, predicting these values remains a significant challenge in the field of Music Information Retrieval. Music is inherently multi dimensional, a high energy track could be a joyful dance anthem (high valence) or a chaotic heavy metal song (low valence). Furthermore, the relationship between audio features and emotional perception is often non linear, requiring sophisticated machine learning techniques to achieve high accuracy.

Research Questions

The goal of this project is to explore the relationship between various Spotify audio features such as tempo (measured in beats per minute, or BPM), loudness, energy, and danceability with music emotions like valence. The research questions that will be answered in this project are as follows

1. To what extent can Spotify's high level audio features accurately predict the valence of a track across diverse genres?
2. Which specific audio characteristics (e.g., danceability) serve as the strongest predictors for positive emotional sentiment?
3. How do non linear machine learning models compare to traditional linear regression in capturing the complexity of musical emotion?

Valence represents the musical positiveness ranging from sad to happy, while Energy (Arousal) measures the physiological intensity and activity level of the track. By predicting these two continuous variables, recommendation engines can provide more context aware suggestions that align with a user's current mood or activity.

2. State of the Art

Recent studies have validated Spotify's proprietary audio features against subjective human ratings, finding that automated metrics are robust for energy/arousal, but that predicting valence remains a distinctly elusive and complex task.

2.1. Psychological Foundations

The computational study of musical emotion is grounded in psychological models of affect. Russell's (1980) Circumplex Model proposed that all emotional states can be described by two independent dimensions: valence (pleasant to unpleasant) and arousal (activated to deactivated). Thayer (1989) adapted this for music, proposing a two-dimensional Mood Meter based on energy and stress. These frameworks became the dominant paradigm for MER.

Eerola & Vuoskoski (2011) conducted a landmark benchmark study comparing dimensional and categorical models of musical emotion. They found that arousal could be predicted with considerably higher accuracy than valence, and that support vector regression and k-nearest neighbours outperformed simpler linear baselines.

2.2. Traditional Machine Learning Approaches

Early computational approaches to MER relied on hand crafted audio features like MFCCs (Mel Frequency Cepstral Coefficients), spectral centroid, tempo, zero-crossing rate combined with traditional machine learning classifiers.

- Yang et al. (2008) used support vector regression, achieving correlation coefficients of approximately 0.64 for arousal but only 0.44 for valence.
- Laurier et al. (2009) combined audio with lyric features, improving valence R^2 by up to 0.08.
- Chaki et al. (2020) applied Random Forest and Gradient Boosting to Spotify features, reporting $R^2 = 0.55$ to 0.75 for energy but substantially lower for valence (0.30 to 0.50).

2.3. Deep Learning and Recent Benchmarks

The past decade saw a shift toward deep learning. Pons et al. (2017) demonstrated that CNNs trained end to end on mel spectrograms outperform hand-crafted approaches for mood classification. Kosta et al. (2022) found that genre stratified models outperform pooled models for valence prediction, with R^2 improvements of up to 0.12 across six genres.

The GlobalMood study (2025) highlights that emotional perception varies cross culturally, necessitating models that handle non-linear relationships. Modern approaches combine audio features with lyrics (sentiment analysis) and facial expression recognition (FER) for real time playlist generation.

Our project situates itself in this landscape by testing whether accessible machine learning models (KNN, Random Forest, XGBoost, Stacked Ensemble) can achieve strong performance on a large industrial dataset with genre stratification into 3 groups with each group having 3 similar genres.

3. Material and Methods

3.1. Dataset

The dataset is the *Spotify Tracks Dataset* sourced from Kaggle.[1] It comprises approximately 114,000 tracks, each described by 21 variables (6 categorical, 11 numerical). There are no significant missing values in the core audio features. The target variables are valence and energy.

Numerical Variables

Variable	Description	Scale / Unit
danceability	How suitable a track is for dancing (tempo, rhythm stability, beat strength, regularity)	0.0 – 1.0
energy	Intensity and activity of a track	0.0 – 1.0
loudness	Overall loudness	dB
tempo	Estimated tempo (speed/pace)	BPM
valence	Musical positiveness conveyed by the track	0.0 – 1.0
acousticness	Confidence measure of whether the track is acoustic	0.0 – 1.0
instrumentalness	Likelihood that the track contains no vocals (values $\rightarrow$ 1.0 = more instrumental)	0.0 – 1.0
speechiness	Presence of spoken words	0.0 – 1.0
liveness	Presence of an audience in the recording	0.0 – 1.0
duration_ms	Length of the track	milliseconds
popularity	Track popularity score	0 – 100

Categorical Variables

Variable	Description
track_genre	Genre to which the track belongs
key	Musical key of the track
time_signature	Estimated time signature
artists	Artist(s) who performed the track
album_name	Album in which the track appears
track_name	Name of the track

3.2. Preprocessing

The following preprocessing steps were applied before modelling:

[1] https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

Column name standardisation. Spaces in column names replaced with underscores for compatibility with R's formula interface.

Type conversion. The `explicit` column was encoded as a binary integer (1 = explicit, 0 = not explicit). Columns `key`, `mode`, and `time_signature` were converted to factors.

Missing value removal. Rows with any missing values removed via `drop_na()`. The original dataset had no missing data, resulting in negligible data loss.

Feature selection. Full predictor set: `popularity`, `duration_ms`, `explicit`, `danceability`, `loudness`, `speechiness`, `acousticness`, `instrumentalness`, `liveness`, `tempo`. Text fields and `key/mode` were excluded.

Feature scaling. For scale sensitive models (Elastic Net, KNN, KNN+PCA, XGBoost), all numeric predictors were z-score standardised using training set statistics only, a critical step to prevent data leakage.

Genre filtering and grouping. Dataset filtered to 9 genres across 3 study groups (9,000 tracks per group). After a 75/25 stratified split: 6,750 training and 2,250 test samples per group.

3.3. Methods

This study applied a diverse set of supervised machine learning models to predict two continuous affective audio attributes

- Valence
- Energy

We are predicting these variables across three genre based subsets of the dataset (Rhythmic, Classical, Rock). All models were implemented in R. The methods used to predict the target variables are:

3.3.1. Linear Regression with Interaction Terms

Linear regression models the relationship between input features and a continuous output:

$$Y = \beta_0 + \beta_1 X + \varepsilon \tag{1}$$

where β_0 is the intercept, β_1 the slope, and ε the residual error. We test five progressively richer predictor sets for valence and energy to understand the marginal contribution of each feature group, ranging from a single predictor to the full 10-feature set.

3.3.2. Elastic Net

Elastic Net combines L1 (Lasso) and L2 (Ridge) penalties:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\left[\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2\right] \tag{2}$$

It adds a penalty that shrinks large coefficients down toward zero. This prevents the model from over relying on any single feature when multiple features are related to each other (for example, loudness and energy tend to move together).

The penalty has a dial called ALPHA:

Alpha = 0.0 means Ridge (shrink all features a little, keep all of them).
Alpha = 1.0 means Lasso (shrink many features to exactly zero, removing them).
Alpha = 0.5 means a 50/50 mix of both approaches.

We tested alpha values from 0.0 to 1.0 and picked the best one for each group and target combination: Both and were tuned simultaneously. For each value of across the grid 0, 0.1, 0.2, . . . , 1.0, the cv.glmnet function in R found the optimal via 5-fold cross-validation with both lambda.min and lambda.1se selection rules. A total of 132 Elastic Net models were fitted.

3.3.3. K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based *lazy learner* predicting outputs as the weighted mean of the K nearest neighbours by Euclidean distance.
To predict the valence or energy of a new song, it finds the K most similar songs in the training data (based on their audio features) and averages their values. It does not assume any straight line relationship, it just looks at what nearby songs are like. Weighted KNN was used with $k \in \{5, 11, 21\}$ via 5-fold cross-validation. Feature scaling is critical for KNN as Euclidean distance is sensitive to feature scale

3.3.4. Random Forest

A Random Forest builds hundreds of decision trees, each trained on a slightly different random sample of the data and using a random subset of features. The final prediction is the average of all the trees. Because each tree is different, the errors tend to cancel out, giving a much more stable and accurate result than any single tree. The final prediction is the mean across all trees. $B = 500$ trees were used, with `mtry` tuned over $\{2, 4, 6, 8\}$.

3.3.5. XGBoost

XGBoost (eXtreme Gradient Boosting) also builds decision trees, but in a very different way from Random Forest. Instead of building all trees independently at the same time, XGBoost builds them one at a time. Each new tree focuses specifically on correcting the errors made by the previous trees. This process is called boosting, each round improves performance by fixing what was wrong before.

Parameters used:
Max tree depth: 6 (how complex each tree can get)
Learning rate: 0.1 (how much each tree is allowed to change the predictions)
Number of rounds: 200 (how many trees are built in total)
Subsample: 0.8 (each tree sees 80
Column subsample: 0.8 (each tree uses 80
Random seed: 42 (so results are exactly the same every run)

3.3.6. Stacked Ensemble

A two-stage meta-learning approach.
Stage 1: five heterogeneous base learners (Linear Regression, Elastic Net, KNN, Random Forest, XGBoost) generate out-of-fold (OOF) predictions via 5-fold cross validation, preventing data leakage.
Stage 2: a non-negative linear meta-learner (no intercept) is trained on the OOF predictions to learn optimal base model weights. Our stacking framework combines five base learners (Linear Regression, Elastic Net, KNN, Random Forest, XGBoost) using 5-fold out-of-fold

(OOF) prediction to generate training data for a non negative linear meta learner without intercept. This design prevents data leakage and ensures the meta learner learns true out of sample generalization.

3.3.7. SHAP Interpretability

SHAP (SHapley Additive exPlanations) is a method for understanding what each audio feature actually contributed to the model's prediction for each song. For each song and each feature, SHAP gives a number that says: "this feature pushed the prediction higher (positive) or lower (negative) by this much."

3.4. Evaluation Protocol

- Data was split 75% / 25% (train / test) with stratified partitioning.
- The three metrics that were computed are:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- 5-fold cross validation for hyperparameter tuning across all approaches to get the best fit and results.

4. Results

4.1. Descriptive Analysis

4.1.1. Overall Statistics

The full analysis dataset comprises 9,000 tracks. Table 2 summarises key distributional properties.

Variable	N	Mean	SD	Min	Q1	Med.	Q3	Max	Skew	Kurt
Valence	9000	0.501	0.271	0.000	0.269	0.507	0.726	0.979	0.000	1.843
Energy	9000	0.612	0.304	0.001	0.373	0.710	0.862	0.998	−0.672	2.106
Danceability	9000	0.526	0.191	0.000	0.385	0.521	0.686	0.957	−0.076	2.166
Loudness	9000	−9.479	7.450	−43.957	−11.574	−6.404	−4.485	−0.163	−1.581	4.765
Speechiness	9000	0.069	0.059	0.000	0.037	0.048	0.074	0.766	3.372	18.134
Acousticness	9000	0.380	0.388	0.000	0.017	0.197	0.804	0.996	0.522	1.599
Instrumentalness	9000	0.164	0.330	0.000	0.000	0.000	0.028	0.982	1.697	4.039
Liveness	9000	0.199	0.179	0.015	0.094	0.126	0.242	0.993	2.349	8.900
Tempo	9000	120.3	33.2	0.000	94.0	114.1	143.1	215.5	0.478	2.375
Popularity	9000	27.3	22.6	0.000	2.0	28.0	40.0	98.0	0.425	2.408

Continued...

Variable	N	Mean	SD	Min	Q1	Med.	Q3	Max	Skew	Kurt
Duration (ms)	9000	228,720	101,173	17,453	181,730	218,964	263,666	2,646,866	6.400	105.7

Table 2: *Descriptive statistics for all 9,000 tracks.*

4.1.2. Genre Group Analysis

We performed analysis per Genre group for valence and energy, and it is represented as Mean SD in the column.

Group	N	Valence	Energy
G1: Rhythmic	3,000	0.696 ± 0.205	0.730 ± 0.132
G2: Classical	3,000	0.303 ± 0.228	0.276 ± 0.245
G3: Hard Rock	3,000	0.504 ± 0.221	0.831 ± 0.157

Table 3: *Valence and energy by genre group (Mean $\pm$ SD).*

G1: Rhythmic Genres (Salsa, Latin, Reggaeton).

- The highest valence is around 0.696 indicating positive and cheerful music.
- The highest energy is around 0.730 and danceability around 0.716
- The data has very low instrumentalness i.e. 0.008 and is predominantly vocal music
- The moderate tempo is around 119.5 BPM.
- The Valence has a range of around 0.047 to 0.979.
- The data for group1 is extremely right skewed.

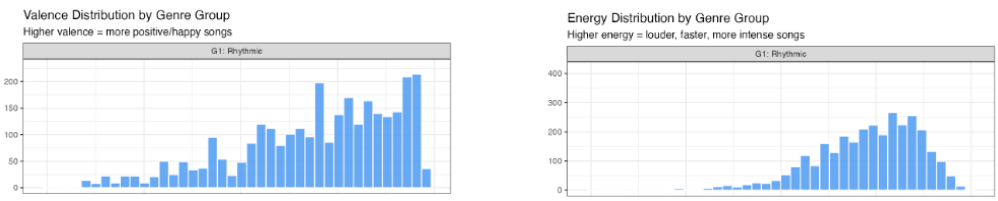

Figure 1: *Valence and Energy Distribution in G1*

G2: Classical Genres (Opera, Classical, Piano).

- The lowest valence is around 0.303 indicating more subdued emotional tone.
- The lowest energy is around 0.276 which shows more calm and peaceful music.
- The Highest acousticness is around 0.807 and has high instrumentalness 0.433 showing heavy use of instruments.
- Negative skewness in acousticness -1.73 that shows most tracks highly acoustic.

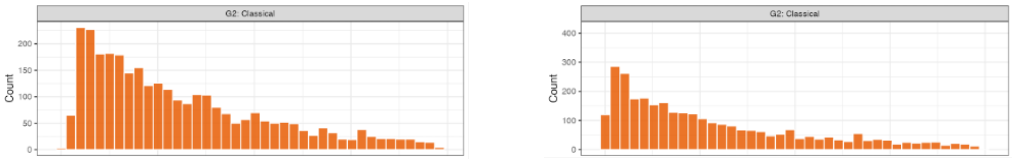

Figure 2: *Valence and Energy Distribution in G2*

G3: Hard Rock Genres (Punk-Rock, Heavy-Metal, Punk).

- The moderate valence is around 0.504 and highest energy is 0.831 intense and powerful music.
- The lowest acousticness is 0.062 and is predominantly electric instruments.
- The data has the most negative energy skewness i.e. -1.43 and is concentrated at high values.
- The data has high liveness around 0.244 suggesting live performance influence

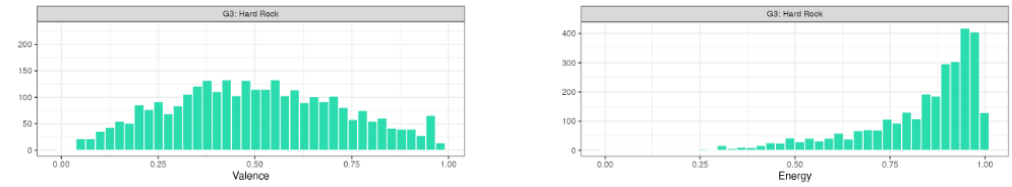

Figure 3: *Valence and Energy Distribution in G3*

4.2. Correlation Analysis

G1 Rhythmic. Valence shows a moderate positive correlation with acousticness at around 0.25 and a weak negative correlation with speechiness, while energy is strongly positively correlated with loudness at 0.63 and moderately negatively correlated with acousticness.

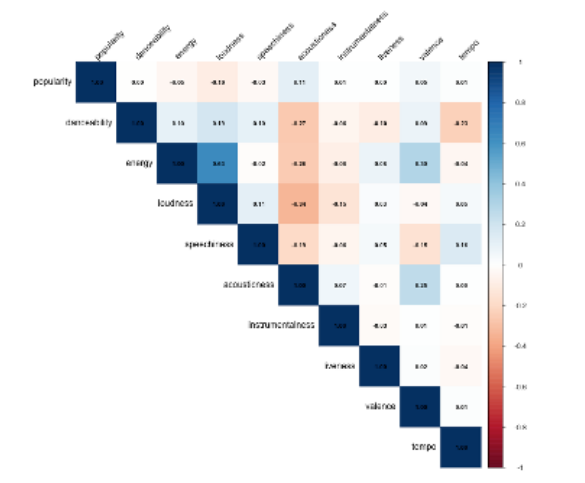

Figure 4: *Correlation G1*

G2 Classical. Valence correlates most strongly with danceability at 0.53, moderately with tempo at 0.23, and negatively with acousticness at -0.23. Energy is very strongly tied to

loudness at 0.83, while acousticness at -0.83 and instrumentalness at -0.58 show strong negative relationships with it.

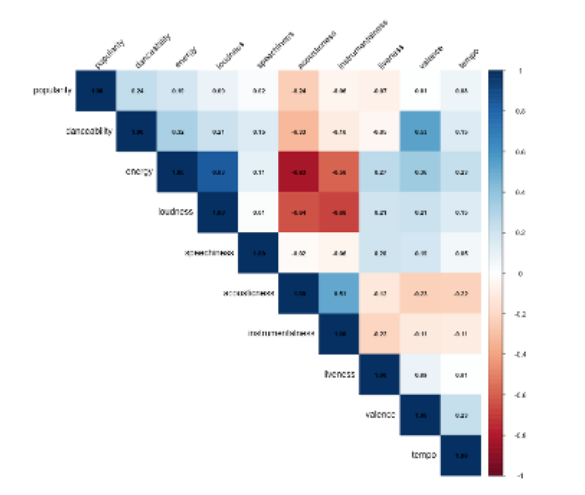

Figure 5: *Correlation G2*

G3 Hard Rock. Valence shows a moderate positive correlation with danceability at 0.40, while energy is strongly tied to loudness at 0.63 and speechiness at 0.36, and strongly inversely related to acousticness at -0.53.

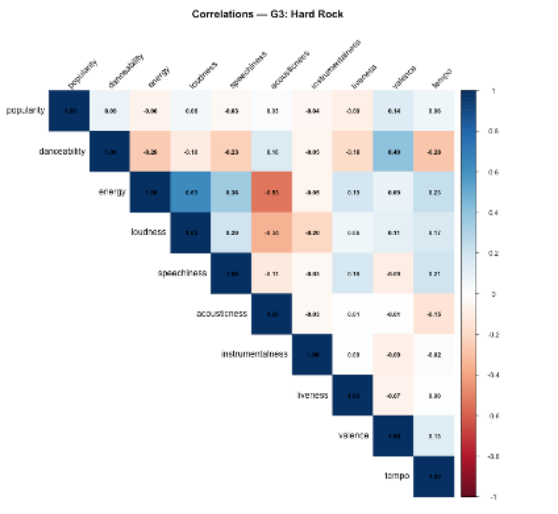

Figure 6: *Correlation G3*

4.3. Linear Model Predictor Set Comparison

Five predictor sets were tested to assess the marginal contribution of each feature group:

M1 Single dominant predictor (Danceability for valence; Loudness for energy)

M2 M1 + two additional features (3 predictors total)

M3 Alternative 3-feature set testing rhythm/acoustic theory

M4 All eight core audio features

M5 M4 + Popularity + Duration (10 predictors)

Set	G1 Energy	G2 Energy	G3 Energy	G1 Val.	G2 Val.	G3 Val.
M1 (1 predictor)	0.4070	0.7192	0.3724	0.0100	0.2830	0.1872
M2 (3 predictors)	0.4176	0.8443	0.5032	0.1075	0.3391	0.2358
M3 (3 predictors)	0.0312	0.2263	0.1416	0.2073	0.3365	0.1973
M4 (8 predictors)	0.5422	0.8741	0.6651	0.3003	0.4066	0.2330
M5 (10 predictors)	0.5680	0.8822	0.6074	0.2944	0.4210	0.2995

Table 4: *Linear Regression Predictors*

Energy is more predictable than valence across all groups. G2 (Classical) has the highest energy R^2 (0.88 with M5), likely because classical music has systematically low energy tied to specific acoustic features (low loudness, low tempo). Valence is harder to predict (max R^2 0.42 for G2 M5), reflecting the subjective and culturally complex nature of perceived musical positiveness. The best valence result with OLS is G2 M5 ($R^2 = 0.421$). Even using all 10 features, a straight-line model can only explain 42 percent of valence variation. Linear regression is a useful starting point and does reasonably well for energy (especially in the Classical group), but it clearly does not show linearity for valence.

Valence proved substantially harder.

4.4. Elastic Net

Elastic Net regression models were trained using 5 fold cross validation, with predictions generated using the λ value that minimised cross validated error (λ_{min}). Most of the best alpha values are low (0.0 to 0.3). This means Ridge style shrinkage works better than Lasso-style feature removal. So, all the audio features carry some useful information, so it is better to keep them all and just reduce their influence slightly, rather than dropping them.

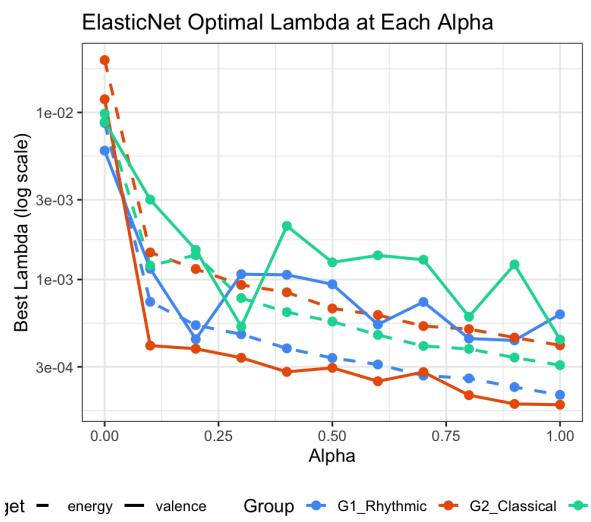

Figure 7: *Optimal Lambda per group*

Elastic Net beats Linear regression M5 for most targets. For G1 energy, RMSE drops to

0.0945 and R^2 goes to 0.4726. The regularisation makes small but real improvements. The biggest valence improvement is for G3: Elastic Net R^2 = 0.3113 vs linear regression M5 R^2 = 0.2995. Consistent improvement. Energy RMSE values are much lower than valence RMSE values across all groups. G2 Energy RMSE = 0.081 means the model is typically off by about 8 points on a 0-1 scale. For G3 Valence, the typical error is 0.184 which is more than twice as large.

Target	Group	Best α	RMSE	MAE	R^2
Valence	G1 Rhythmic	0.2	0.1675	0.1308	0.3257
Valence	G2 Classical	0.1	0.1723	0.1327	0.4228
Valence	G3 Hard Rock	0.3	0.1836	0.1507	0.3113
Energy	G1 Rhythmic	0.0 (Ridge)	0.0945	0.0745	0.4726
Energy	G2 Classical	0.9 (Lasso)	0.0811	0.0613	0.8842
Energy	G3 Hard Rock	0.1	0.1004	0.0755	0.6088

Table 5: *Elastic Net best results (λ_{min}).*

4.5. Comparing KNN and KNN with Dimentionality Reduction(PCA) results

PCA preprocessing provides no consistent benefit for KNN across groups.

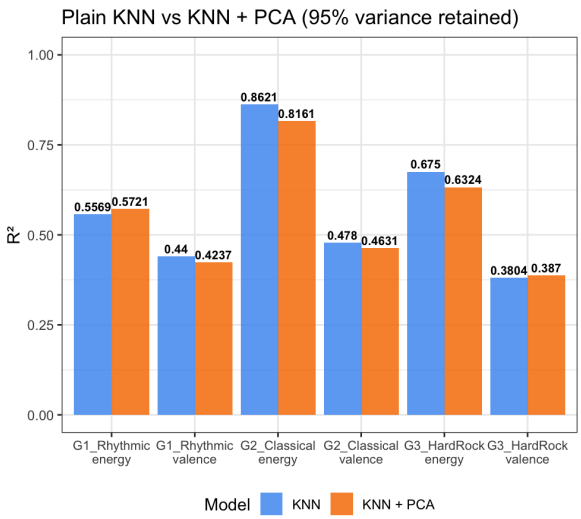

Figure 8: *PCA VS PCAKNN*

- In 4 of 6 cases PCA hurts performance, because the original feature space is already low dimensional (10 features) and PCA induced compression discards discriminative variance.
- The two cases where PCA helps (G1 energy +0.015, G3 valence +0.007) show only marginal gains, further supporting the conclusion that raw features are preferable.

- For Valence, the data showing non-linear relationships and PCA doesn't help cannot help when the data is already non linear

Group	Target	KNN R^2	KNN+PCA R^2
G1 Rhythmic	Valence	0.4400	0.4237
G1 Rhythmic	Energy	0.5569	0.5721
G2 Classical	Valence	0.4780	0.4631
G2 Classical	Energy	0.8621	0.8161
G3 Hard Rock	Valence	0.3804	0.3870
G3 Hard Rock	Energy	0.6750	0.6324

Table 6: *KNN vs. KNN + PCA comparison* (R^2).

4.6. Tree Based Method Analysis: Random Forest

Random Forest is the biggest improvement in this study compared to the linear models. For G1 Valence, the R^2 goes from 0.326 (Elastic Net) to 0.600 (Random Forest) nearly doubles the explained variance. This is because Random Forest can pick up on curved patterns and combinations of features (for example: "if danceability is high AND acousticness is low, valence tends to be higher") that linear models were missing.
Random Forest is the strongest single model in this study. It handles the non-linear nature of the data well and produces stable, reliable results.It is the main comparison point for all other models.

Group	Target	RMSE	MAE	R^2
G1 Rhythmic	Valence	0.1325	0.0877	0.5997
G1 Rhythmic	Energy	0.0687	0.0451	0.7172
G2 Classical	Valence	0.1487	0.1091	0.5683
G2 Classical	Energy	0.0705	0.0503	0.9156
G3 Hard Rock	Valence	0.1465	0.1094	0.5634
G3 Hard Rock	Energy	0.0656	0.0418	0.8262

Table 7: *Random Forest performance.*

4.7. Tree Based Method Analysis: XGBoost

XGBoost's biggest strength in this study is G2 Energy, where it achieves the highest R^2 among all single models: 0.9171. Classical music energy has very consistent patterns like loudness, tempo, and acousticness together define it almost perfectly and XGBoost is excellent at exploiting such patterns.

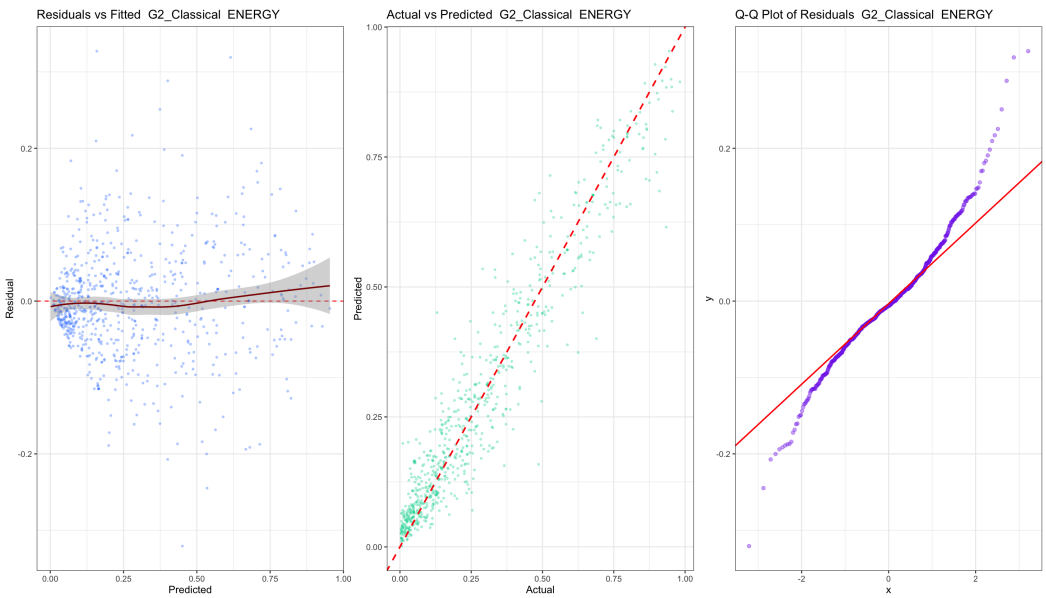

Figure 9: *G2: Classical best performing XGboost Plots*

4.8. Stacked Ensemble Analysis

The stacked ensemble consistently improves or matches the best base model in 5 of 6 cases. It corrects systematic errors that no single model avoids. Random Forest already achieves R^2=0.826, the meta learner introduces slight variance when a single strong learner has near-optimal fit. Overall, stacking provides reliable marginal improvements for valence prediction where individual models leave more residual variance to exploit. Stacking is the

Group	Target	Best Base Model	Base R^2	Ensemble R^2	Gain
G1 Rhythmic	Valence	Random Forest	0.5997	0.6029	0.0032
G1 Rhythmic	Energy	Random Forest	0.7172	0.7207	0.0035
G2 Classical	Valence	Random Forest	0.5683	0.5792	0.0109
G2 Classical	Energy	XGBoost	0.9171	0.9186	0.0015
G3 Hard Rock	Valence	Random Forest	0.5634	0.5727	0.0093
G3 Hard Rock	Energy	Random Forest	0.8262	0.8219	–0.0043

Table 8: *Stacked Ensemble vs. best base model* (R^2).

recommended final model for most targets in this study. It reliably improves over single models at low extra cost. The only exception is G3 Energy, where Random Forest alone should be used as it is already the best possible result.

4.9. Overall Valence Performance

Every step up in model complexity brings a real improvement. Going from Linear Regression (R^2 = 0.07 for G1) to Stacked Ensemble (R^2 = 0.60 for G1) is a massive gain. However, most of that gain comes from the jump to tree based models (Random Forest and XGBoost). Elastic Net and KNN each add moderate improvements over linear regression, while tree models

make the biggest jump.

Model	G1 RMSE	G1 MAE	G1 R^2	G2 RMSE	G2 MAE	G2 R^2	G3 RMSE	G3 MAE	G3 R^2
Linear Regression	0.2018	0.1602	0.0704	0.1800	0.1378	0.3670	0.1913	0.1566	0.2557
Elastic Net	0.1675	0.1308	0.3257	0.1723	0.1327	0.4228	0.1836	0.1507	0.3113
KNN	0.1567	0.1074	0.4400	0.1635	0.1207	0.4780	0.1745	0.1392	0.3804
KNN + PCA	0.1589	0.1098	0.4237	0.1658	0.1237	0.4631	0.1736	0.1361	0.3870
Random Forest	0.1325	0.0877	0.5997	0.1487	0.1091	0.5683	0.1465	0.1094	0.5634
XGBoost	0.1442	0.1038	0.5257	0.1487	0.1109	0.5681	0.1550	0.1192	0.5114
Stacked Ensemble	0.1319	0.0885	0.6029	0.1468	0.1077	0.5792	0.1449	0.1058	0.5727

Table 9: *Full model comparison Valence prediction.*

G2 Classical valence (R^2 up to 0.579). Classical and acoustic songs follow more predictable emotional patterns a slow, quiet piece tends to score low on valence consistently.
G3 Hard Rock valence is the hardest (best R^2 = 0.573). Fast, loud metal tracks can be angry (low valence) or triumphant (high valence), and audio features alone cannot always tell the difference.
G1 Rhythmic valence top R^2 = 0.603 (Ensemble). Dance and pop songs vary a lot in valence and the audio features provide a moderate but imperfect signal.

4.10. Overall Energy Performance

Even the simplest model (Linear Regression) does very well for Energy. Classical music energy is so closely tied to loudness and tempo that a straight line already captures most of the pattern.

Model	G1 RMSE	G1 MAE	G1 R^2	G2 RMSE	G2 MAE	G2 R^2	G3 RMSE	G3 MAE	G3 R^2
Linear Regression	0.0966	0.0764	0.4414	0.0897	0.0665	0.8634	0.1017	0.0746	0.5818
Elastic Net	0.0945	0.0745	0.4726	0.0811	0.0613	0.8842	0.1004	0.0755	0.6088
KNN	0.0860	0.0599	0.5569	0.0901	0.0644	0.8621	0.0897	0.0601	0.6750
KNN + PCA	0.0845	0.0585	0.5721	0.1041	0.0752	0.8161	0.0954	0.0638	0.6324
Random Forest	0.0687	0.0451	0.7172	0.0705	0.0503	0.9156	0.0656	0.0418	0.8262
XGBoost	0.0723	0.0530	0.6865	0.0699	0.0505	0.9171	0.0684	0.0451	0.8109
Stacked Ensemble	0.0683	0.0455	0.7207	0.0692	0.0495	0.9186	0.0664	0.0422	0.8219

Table 10: *Full model comparison Energy prediction.*

G2 Energy results converge near the top: Elastic Net (0.884), KNN (0.862), Random Forest (0.916), XGBoost (0.917), Ensemble (0.919) all very high, and the gap between models shrinks because there is less room for improvement once the main patterns are learned.

G1 Energy is harder than G2 (best R^2 = 0.721). Rhythmic genres mix electronic and acoustic styles, fast and slow songs, all under the same "rhythmic" label. Energy varies in more complex ways.
G3 Energy best single model is Random Forest (R^2 = 0.826, RMSE = 0.066). This is the one case where the Ensemble (R^2 = 0.822) does not beat the best single model. Random Forest has already found the best fit for G3 Energy.

4.11. SHAP: Understanding what drives prediction for each Groups

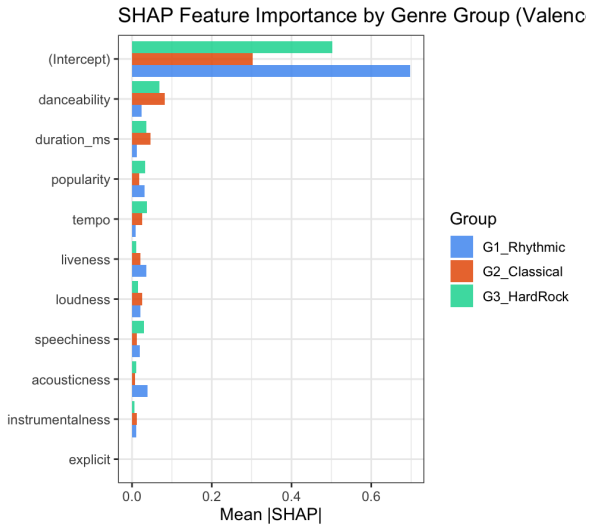

Figure 10: *SHAP Feature Imprtance*

- Danceability matters most in G2 Classical in that group, a track's rhythmic quality is a strong anomaly signal, predicting when present.
- Acousticness influences more in G1 Rhythmic, tracks that are acoustic are strongly distinguished from their electric counterparts.
- Tempo drives G3 Hard Rock energy, consistent with the genre's use of fast BPM as an intensity marker.
- Liveness is very influencial for G1 (live performance recordings have notably higher energy in dance/pop contexts) but near zero in G3.

Feature	G1 SHAP	G2 SHAP	G3 SHAP	Divergence	Most Important In
Danceability	0.0245	0.0822	0.0691	0.0577	G2 Classical
Duration (ms)	0.0121	0.0461	0.0355	0.0340	G2 Classical
Acousticness	0.0385	0.0068	0.0109	0.0317	G1 Rhythmic
Tempo	0.0098	0.0252	0.0369	0.0271	G3 Hard Rock
Liveness	0.0366	0.0213	0.0104	0.0262	G1 Rhythmic
Speechiness	0.0191	0.0123	0.0306	0.0183	G3 Hard Rock
Popularity	0.0315	0.0174	0.0324	0.0150	G3 Hard Rock
Loudness	0.0215	0.0256	0.0155	0.0100	G2 Classical
Instrumentalness	0.0113	0.0126	0.0057	0.0069	G2 Classical
Explicit	0.0000	0.0000	0.0000	0.0000	None

Table 11: *Mean absolute SHAP importance Valence prediction.*

Key Takeway from Shap:
Different audio features matter in different genre groups. This is the core reason why genre stratified models work better than one big model. If we trained one model on all 9,000 songs, it would be forced to average these very different patterns, making it worse for all groups.

Summary of genre specific feature importance:
G1 Rhythmic: Acousticness and Liveness are the key features.
G2 Classical: Danceability and Duration carry the most useful information.
G3 Hard Rock: Tempo and Speechiness are influential.

However, the SHAP analysis suggests that genre level differences in average valence are a dominant factor.

5. Discussion

5.1. The Valence Prediction

The results demonstrates the difficulty of predicting valence from audio features. Even the best models achieve $R^2$0.57 to 0.61 for valence, compared to 0.59 to 0.92 for energy. This valence and energy asymmetry is consistent across all genre groups and model families.

The difficulty in valence prediction can be attributed to four compounding factors:

1. **Subjective interpretation** Valence is a fundamentally listener dependent construct, the same audio pattern can be perceived as joyful or melancholic depending on cultural and personal context.
2. **Cultural context** The emotional meaning of musical features is genre and culture dependent, as confirmed by the divergent SHAP profiles observed here.
3. **Missing lyrical features** Lyrical content is absent from the feature set, prior work (Laurier et al., 2009) shows lyrics can improve valence R^2.
4. **Genre specific patterns** No single feature is consistently dominant for valence across all groups, making a universal predictor elusive.

5.2. Non Linearity and Model Complexity

Results clearly demonstrate that the feature emotion relationship is substantially non linear for valence. For energy, the degree of non linearity is group dependent: G2 Classical energy is nearly linear (loudness alone explains ~72% of variance), while G1 Rhythmic energy exhibits a much larger non linear component.

Importantly, increasing model complexity does not always improve performance. KNN+PCA consistently reduces performance, and adding metadata features hurts G3 Hard Rock energy prediction. In low-dimensional feature spaces ($p = 10$), dimensionality reduction is counterproductive and feature selection matters more than raw model complexity.

5.3. Genre Stratification

The dramatically different SHAP profiles across groups, acousticness and liveness dominating G1, danceability dominating G2, tempo and speechiness dominating G3 would be completely obscured in a pooled model. This finding validates the genre stratified design and music emotion recognition systems should be genre aware rather than genre agnostic.

Practically, this implies production recommendation systems should route tracks to genre specific emotion models rather than relying on a single universal model.

6. Conclusion

This study presented a comprehensive genre stratified machine learning analysis of valence and energy prediction from Spotify audio features across 9,000 tracks and three genre groups. The key findings are:

- **Energy is reliably predictable** from audio features as energy is a physical property of sound (loudness, tempo, distortion) that audio analysis can measure directly. Valence is an emotional quality.
- **Valence prediction is challenging** however, non linear models provide substantially better results than linear baselines.
- **Random forest and XGboost are both excellent** Random Forest wins in 5 of 6 direct comparisons. XGBoost wins for G2 Energy. In practice, both models produce very similar results. Random Forest handles noisy valence data slightly better, while XGBoost excels at structured, consistent energy patterns like classical music.
- **The Stacked Ensemble is the most consistent top model**, winning 5 of 6 group-target comparisons. For valence, there is a very large gap between linear models (R^2 around 0.07 to 0.42) and tree-based models (R^2 around 0.51 to 0.60). The feature to valence relationship involves curves and combinations of features that require tree models to detect.
- **Splitting by Genre group is essential**. The three groups follow very different rules. A single model trained on all 9,000 songs would be confused by these differences and perform worse for everyone. Separate models for each group are clearly the right approach, and the SHAP analysis confirms that different features drive each group.
- **PCA preprocessing does not help KNN** in this low-dimensional ($p = 10$) feature space and consistently hurts performance.
- **Feature Importance differs by Genre** There is no single most important audio feature for predicting emotions. Which feature matters depends on the genre group. Acousticness is key for Rhythmic (G1), Danceability is key for Classical (G2), Tempo is key for Hard Rock (G3). This genre specific finding is one of the most interesting conclusions of the study.

7. Future Work

- Lyrical features via sentiment analysis or pre trained language model embeddings. Words are the most direct carrier of emotional meaning in music. The emotions can be different even if both have identical audio features. So, working on sentiment analysis as an extention of the project can be very useful.
- Raw audio deep learning representations (e.g., mel spectrogram CNNs). Spotify's audio features (danceability, loudness, tempo, etc.) are pre computed summaries. A more direct approach is to work from the raw audio signal itself using deep learning.
- Listener demographic and contextual data to personalise emotion predictions. The same

song can feel happy or sad depending on who is listening and when. These listener side effects are called contextual or personalized emotion.

- Cross-cultural validation to assess the generalisability of learned feature emotion mappings. This study was conducted on Spotify data that is heavily weighted toward Western popular music genres (pop, rock, EDM, classical). The learned relationships between audio features and emotion may not transfer to music from other cultural traditions.

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Github Link

https://github.com/hrithik99-afk/PredictionEmotioninMusic